

COURTSIDE: trading the seconds after a tennis point

Gator Quant Hacks 2026 · Systematic Trading track · Reproduce: `bash reproduce.sh` · Code: github.com/OjasMishra32/computervisionpredictivetabletennismoneymachineouuushiii

Summary

Strategy (v2): take the stale quotes left on Polymarket tennis moneylines in the 0–3 s after a point, sized by risk, capped at 100 net shares per match, held to resolution. **Edge:** a persistent *fast tier* of traders who know the point before the book reprices is paid by everyone slower.

Out of sample: the fast tier beat the market in 3 of 3 held-out months while other takers lost. Our first strategy (v1) lost \$36k on the held-out window, opened once, blind. v2 was designed after that, so its held-out result is non-blind: +0.60¢/share [0.09, 1.13], Sharpe 6.7, gone at half a tick of slippage and **negative when costs double**. On 11,307 never-examined markets v2 failed its pre-registered out-of-sample test. The blind forward test is pending (§4). The edge is real at the fast tier's speed; v2 is not proven.

1. Economic hypothesis

A tennis point ends in tiers. Ball tracking knows where the ball lands while it is in the air, the umpire at the bounce, licensed data feeds (and the market makers on them) about a second later, and streams and score widgets tens of seconds later. Win probability is a known function of the score (our exact point-level Markov chain, `src/markov.py`): a simulated ATP best-of-3 has ~161 points, mean move 5.7%.

H6. We expect Polymarket ATP/WTB moneylines to trade at stale prices for 1–3 s after each point, because takers on slower tiers trade against quotes that faster tiers already know are wrong. The edge persists because the gaps are physical (frame rates, data licensing, stream delays) and the venue holds every marketable sports order for 1 s (3 s before May 2026) to protect makers. If true, wallets trading within 3 s of a detected score event earn positive net 30 s markouts month after month while other takers lose. It fails if the month-m net markout is ≤ 0 in more than a third of months, or ≤ 0 out of sample.

Who is on the other side: whoever acts on an older tier. The edge is structural (latency), not a risk premium, and it responds as the theory predicts: in sample the fast tier's edge per share fell as the venue's protection fell (1.51¢ at a 3 s delay and no fee, 1.64¢ at 3 s/3%, 1.19¢ at 1 s/3%, 0.54¢ at 1 s/5%) and as qualifying wallets grew from 4 to 131, yet stayed positive every month. **What is new:** courtsiding and latency arbitrage are old ideas and the tennis Markov model is standard (Klaassen & Magnus 2001). We add a walk-forward, wallet-level identification of the latency tier from public on-chain tapes, a timing of every information source against the official point clock, and how early ball tracking can call a point. H1–H4 were committed before any result (HYPOTHESIS.md, commit 7232986); H5–H6 after in-sample results, frozen before the OOS; v2 in HYPOTHESIS_V2.md; every change in DEVIATIONS.md.

2. Data and universe

- **Instruments, frequency, dates.** Every resolved ATP/WTB (incl. Challenger) singles moneyline on Polymarket with $\geq$ \$5k volume, Oct 8 2025 – Oct 3 2026: 13,084 matches, \$2.84B traded. Binary contracts on a public order book, with free tick data. Each taker print is classified as lifting the ask or hitting the bid, so fills pay the spreads actually traded. Tape timestamps are block times, a median 1.98 s after the true trade time (5,472 trades matched by transaction hash).
- **Sources** (see References): Polymarket Gamma, Data and CLOB APIs and sports websocket; Kalshi API; ESPN scoreboard and WTA API (latency only); Ken French Data Library;

OpenTTGames and TrackNet video; BlurBall/WASB-SBDT detector weights.

- **Holdout.** In sample: the first 80% of matches by start time. Out of sample (OOS): the last 20%, 2,617 matches from 2026-08-25 14:15 UTC (21% of volume; the 2-year option does not bind). Volume grew over the year, so this covers only 11% of the calendar span; a 20%-of-time split would start on Jul 23. Forward: matches starting $\geq$ 2026-10-03 14:00 UTC, the blind test for v2.
- **Survivorship, corporate actions, missing data.** The universe is the venue's full event list, winners and losers, including the 2.9% of matches settled 50/50. But the \$5k filter uses lifetime volume, known only after the match: it drops thin markets a live trader would see (our check is the blind test on 11,307 smaller markets, §4). Binary contracts have no corporate actions. All 13,084 tapes were fetched; 6 reach the data API's offset cap and may be truncated (kept, not filled). The 2,889 matches with no published fee schedule (all before Apr 2026) are charged 0%, the venue's early no-fee regime.

3. Methodology

- **Signal.** A detector fires when the 10 s volume-weighted price moves $\geq$ 4¢ from the previous 60 s. Each month m , a wallet joins the fast tier if, on earlier months only, it has $\geq$ 30 prints within 0–3 s of a detection over $\geq$ 10 matches and $t > 3$ on its 30 s markout. v2 keeps a wallet only if its shrunk (no = 200) past markout beats the fee in force at that price.
- **Portfolio, sizing, rebalancing.** v2 takes each qualifying print inside 0–3 s of detection with price q in $[0.05, 0.95]$. Shares $\propto 1/\sqrt{q(1-q)}$ (risk parity), never above the copied print or \$1k. Net exposure $\leq$ 100 shares per match; risk-reducing trades always allowed. Event-driven ($\sim$ 270 trades a day), held to resolution: no exits, no rebalancing.
- **Execution.** v2 fills at the fast tier's own print prices: it prices the opportunity for a trader as fast as the fast tier, not our execution today (§6). No decision uses future data: wallets qualify on past months, the fee filter uses the fee in force, and the window starts when the detector could fire (D9). The executable lagged version (the same prints copied 3 s later at mid plus half the spread) loses every month (Table 1). Table 2 also charges $\frac{1}{2}$ and 1 tick of worse entry.
- **Costs.** Each match's own taker fee, $\text{rate} \cdot q(1-q)$ per share = $10,000 \cdot \text{rate} \cdot (1-q)$ bps of notional (250 bps at $q = 0.5$ at today's 5%). On v2's trades it averages 120 bps in sample (0, 3 and 5% regimes) and 179 bps out of sample. Spreads are paid as traded; stress tests add half a 1¢ spread ($\approx$ 95–99 bps).
- **Validation.** Monthly walk-forward inside the in-sample period. Labels are 30 s markouts, so they overlap the next fold by $\leq$ 30 s and we used no further purge gap. v2's sizing and net cap were chosen on the whole in-sample period (PBO, §4).

4. Results

Table 1. Tests, net of fees and traded spreads (¢/share; 95% CI clustered by match).

Test	In sample	OOS (opened once)	Verdict
H1 Follow the jump after the delay	−1.61 [−1.65, −1.57]; all 20 variants < 0	−2.08 [−2.17, −1.98]	Fails
H2 Buy favourites entering 0.85–0.97	−0.41 [−1.44, 0.59]; calibrated $\sim$ 1¢	+0.80 [−1.36, 2.66]	No edge
H5 Quote after jumps (post-hoc)	+0.22 [−0.01, 0.43]	+0.16 [−0.49, 0.87]	Inconclusive
H6 Fast tier (post-hoc)	+0.6 to +2.4; 8/8 months > 0	+0.4 to +0.8; 3/3 > 0	Holds

Test	In sample	OOS (opened once)	Verdict
Everyone else, same 0–3 s window	−0.5 to −1.7, every month	−1.2 to −1.9	
Copying the fast tier 3 s later	< 0 every month	< 0 every month	Speed

v1 (the fast tier's trades, $\leq$ \$1k each, held to resolution) earned +1.16¢/share in sample but **lost \$36k out of sample**: big tickets on cheap tokens made its P&L a lottery. Six optimisation lenses then ran on in-sample data only, each attacked by two adversarial verifiers (research/v2/). Their survivors are v2, which was designed knowing how v1 failed, so the held-out window is **burned** for v2.

Table 2. v2, causal window, net of fees, held to resolution. Capital = $3 \times$ peak locked.

	In sample, Feb 1–Aug 25 (206 d)	Burned OOS, Aug 25–Oct 3 (40 d)
Trades	55,662	10,412
Net ¢/share at fast-tier fills	+1.38 [1.17, 1.59]	+0.60 [0.09, 1.13]†
... +½ tick / +1 tick worse entry	+0.88 / +0.38	+0.10 / −0.40
... fees $\times 2$	+0.77 [0.57, 0.98]	−0.34 [−0.86, 0.19]
... all costs $\times 2$	+0.27 [0.07, 0.48]	−0.84 [−1.36, −0.31]
Net edge per \$ traded	273 bps	114 bps
P&L / capital	+\$40.4k / \$28.3k	+\$3.7k / \$22.8k
Annualised return / volatility	253% / 17.5%	148% / 22.2%
Sharpe / max drawdown	14.5 / −2.0%	6.7 / −2.1%
Skew / worst day / worst month	0.53 / −1.9% / +\$2,543	0.33 / −2.1% / −\$434
Turnover per year	93 $\times$	129 $\times$
Months positive	7/7 (7/7 at +1 tick)	2/3 (Oct = 3 days)

† Clustered by copied wallet instead: [−0.60, 2.11]; out of sample a handful of the fastest wallets carry the profit. All costs $\times 2$ = fee twice plus half a 1¢ spread. Returns are arithmetic, on capital with a 4 h lock per position (realised lock times need \$34.8k in sample, docs/RISK.md); 40-day figures are short-sample. Derived rows: scripts/note_metrics.py.

Costs doubled: the out-of-sample edge disappears. With fees doubled, held-out v2 loses 0.34¢ a share (0 of 3 months positive); with all costs doubled, 0.84¢. In sample it survives both (7/7 months; Sharpe 3.0). Today (1 s delay, 5% fee, 131 qualifying wallets) v2 pays only whoever is first to the stale quote; the margin for second place is under half a tick.

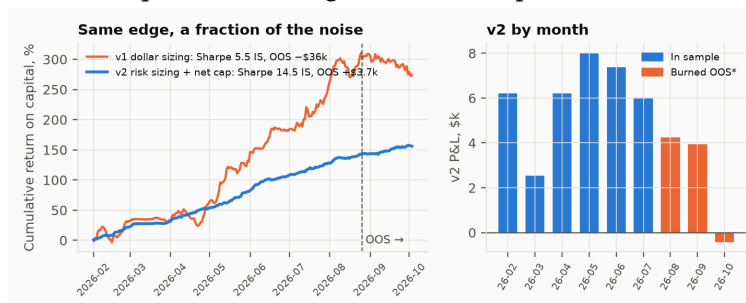


Fig. 1. Equity curve: v1 vs v2 return on each book's own capital; v2 P&L by month (*Aug includes in-sample days).

Sharpe above 3. We assumed a bug and found one: onset labelling in hindsight put 81% of a v2 draft's OOS P&L before detection (D9); the fix cut in-sample Sharpe from 16.8 to 14.5. What remains: ~ 270 small, nearly independent bets a day on exogenous binary outcomes, with net exposure capped per match. All nine net-cap and deployment settings in the sizing

study have per-share CIs above zero (a plateau, not a spike); the probability of backtest overfitting is 0% on the 55-policy sizing grid and 9–24% on the 24-variant v2-safe grid. **Luck:** the deflated Sharpe at $N = 3,386$ trials is 0.997 in sample. Over 40 days the bar is far higher: the expected best of 3,386 zero-skill trials is a Sharpe of 11.0, above v2's 6.7 (DSR 0.075; 0.48 at $N = 44$; $P(\text{Sharpe} > 0) = 0.987$; block-bootstrap CI [1.9, 12.3]). **40 days cannot rule out luck for v2.**

Blind test on 11,307 never-examined markets (pre-registered; mostly ITF $\geq$ \$1k volume; research/v2/expand/): v2 earned +2.02¢ [1.14, 2.93] in the in-sample period (pass) and +1.22¢ [−0.19, 2.65] out of sample, a **fail** by our rule. The mechanism held: fast-tier wallets beat all other takers by +3.09¢ [2.89, 3.28] in sample and +2.16¢ [1.79, 2.53] out of sample per print.

Forward test (blind; matches starting $\geq$ 2026-10-03 14:00 UTC; run once; primary A, fast tier minus other takers, and primary B, the v2 book, both 30 s net, each pass only if the match-clustered 95% CI excludes 0): **pending at the time of writing** (results/v2/forward.json).

Every look at held-out data (results/oos_peeks.log, 19 lines): v1 once, blind; first runs of three pre-registered blind tests (v2 and v2-safe on unseen markets, a side-market maker); 12 non-blind reads of the burned window (v2 $\times 2$, v2-safe, cost stress, a capacity curve, tier-0 counterfactual $\times 7$); one re-run reproducing a committed result; two descriptive reads of forward-window order books (no strategy or P&L). Rules changed after a look twice: v2 itself, and the D9 fix (held-out v2 +0.75¢ $\rightarrow$ +0.60¢).

5. Risk management

- **Limits (backtested; full register in docs/RISK.md).** $\leq$ \$1k per order; $|\text{net exposure}| \leq 100$ shares per match; price zone 0.05–0.95; peak dollars locked across all open matches \$9.4k. Worst match −\$206 (v1: −\$3,098). No per-wallet or cross-match gross cap was tested.
- **De-risking (set in advance; not backtested).** Re-price the wallet filter at the live fee monthly. Halve size if the trailing-month net edge is below 0.3¢; stop at ≤ 0 . Trade tracking calls only at $P \geq 0.95$. The paper engine (engine/risk/) adds a \$1,000 daily stop ($\approx 2 \times$ the worst in-sample day) and kill switches on a stale feed (> 2 s) or tracking (> 1 s), or feed delay above its 95th percentile.
- **Venue rules, the largest risk.** The move to a 5% fee cut the fast tier's edge by about two-thirds; doubling today's fee erases the held-out edge (Table 2). A longer order delay would cut it again.
- **Crowding and concentration.** Wallets grew 4 $\rightarrow$ 131 and edge per share fell from ~ 2.4 ¢ to ~ 0.8 ¢, still positive every month. Out of sample the top five copied wallets carry 98% of the blind-test profit; monthly requalification with shrinkage is the only guard against their exit.
- **Tail and regime.** Kurtosis 4.3 / 3.3 (in / out of sample), worst day −1.9% / −2.1%. Risk dial v2-safe (net cap 50, chosen in sample, pre-registered): burned-OOS worst day −\$224 vs −\$469, Sharpe 9.3 vs 6.7, yet it still fails the blind test. Caps shrink losing days, not their number.
- **Factor exposure.** On Fama–French market, size, value and momentum (142 days): alpha +0.79%/day ($t = 8.5$), every beta insignificant (largest $t = 1.33$), $R^2 = 3\%$.
- **Settlement and legal.** 2.9% of matches settled 50/50 (mostly walkovers), included in all P&L. Courtsiding breaks most ticket terms, tracking data is licensed and Polymarket's international venue restricts US persons, so deployment needs licensed data, a permitted venue and legal review. We never place orders.

6. Liquidity and capacity

- **Depth and impact.** Median spread 1¢, \$8.1k at the touch, \$61k within 2¢ (live sample); table tennis is untradable (89¢ spread, \$23 at the touch). v2 never takes more than the print it copies, so the limit is stale depth per point: median \$222–565 rests at prices the next reprice makes stale, gone 0.5 s after it (482 live points). Costs and slippage: \$3 and Table 2.
- **Capacity.** v2 traded \$1.48M in 206 days (~\$7.2k a day, 0.07% of the same matches' volume) on \$28k of capital. The sizing lens's frontier (1 s/5% regime, per 30 days, onset labels): \$23k of capital trades \$259k at Sharpe 16.8, \$79k trades \$1.10M at 6.6, \$102k trades \$1.65M at 6.0. So roughly **\$100k** before Sharpe falls to ~6; beyond that, growth means displacing the fast tier (\$0.3–3.1M a month in the window).
- **Latency: what deployment needs.** The book reprices 1.2 s *before* the official WTA point stamp (482 points); ESPN, Polymarket's sports feed and the WTA API trail by 27.5, 29.1 and 43.3 s. With the venue's 1 s delay, an order must leave ≥ 1.3 s before the reprice, ≈ 2.5 s before the stamp. Prints in the 0.5 s before a reprice were 98% with the move, so some takers have the point 1–1.5 s before the book. Where such a lead can come from: simulated Hawk-Eye-class tracking (assumed 340 fps, ± 3.6 mm) lands within ± 2.4 cm 100 ms before the bounce; real 120 fps video (OpenTTGames) called 11 of 11 table-tennis misses correctly 50 ms before contact (recall 27%); 25 fps broadcast tennis (TrackNet) is useless. Co-locating near Polymarket's origin saves ~130 ms per round trip.

7. Limitations and next steps

- **What did not work** (all reported): chasing the move (H1), favourite bias (H2), quoting after jumps (H5), maker exits (adversely selected on live books), side-market sniping (~\$2k/month), the Kalshi→Polymarket laggard (≈ 0 net), v1, and a side-market leaning maker (blind OOS +1.87¢ [$-0.14, 3.86$], a fail; [research/v2/maker/](#)).
- **Variants tried: 3,410** (44 for H1–H6; 3,342 across six v2 lenses: exit 902, sizing 111, selection 341, cross-market 68, latency 16, Kalshi 1,904; 24 for v2-safe), plus 432 tier-0 counterfactual scenarios on assumed latency inputs, 4 cost-stress cases and size-scaled capacity runs, none used to choose v2. All are in [results/](#) and [research/](#), with verifier reports.
- **What could break it:** a higher fee or longer order delay, faster rivals at the front of the queue, the few fastest wallets leaving, venue access and legal limits, and our own distance from tier-0 speed.
- **With more time:** run the forward test for weeks; build the tier-0 path end to end (in-venue tracking → fair value → co-located order) and measure real fills; test per-wallet caps; trade on Kalshi.

References. Bailey & López de Prado (2012) J. Risk; (2014) *The deflated Sharpe ratio*, JPM. Bailey, Borwein, López de Prado & Zhu (2017) *The probability of backtest overfitting*, J. Comp. Finance. Harvey, Liu & Zhu (2016) RFS. Politis & Romano (1994) JASA. Klaassen & Magnus (2001) JASA. Fama–French (1993) and Carhart (1997) factors, Ken French Data Library. Voeikov et al. (2020) TNet/OpenTTGames; Huang et al. (2019) TrackNet; Gossard et al. (2026) BlurBall; Tarashima et al. (2023) WASB-SBDT. Polymarket Gamma, Data, CLOB and sports-feed APIs and fee docs; Kalshi API; ESPN scoreboard; WTA API. numpy, pandas, scipy.